

Class17 HW: RNA-Seq Galaxy

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Libraries

```
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.1      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.2.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggplot2)
library(readr)
```

Data Import

```
galaxy_data <- read.table("expression_results.txt")
```

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
galaxy_data |>
  group_by(geno) |>
  summarize(
    Sample_Size = n(),
    Median_Expression = median(exp)
  )
```

```
# A tibble: 3 x 3
  geno Sample_Size Median_Expression
<chr>      <int>          <dbl>
1 A/A         108           31.2
2 A/G         233           25.1
3 G/G         121           20.1
```

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

The SNP seems to effect the expression of ORMDL3. Individuals with the A/A genotype show higher expression levels compared to those with the G/G genotype. This suggests that the presence of the A allele may be associated with increased expression of the ORMDL3 gene; or that the G allele could be linked to decreased expression. This could have additive effects if an individual is heterozygous (A/G), resulting in intermediate expression levels as seen below.

```
ggplot(galaxy_data) +
  geom_boxplot(aes(x = geno, y = exp, fill = geno, notch=TRUE)) +
  labs(
    title = "Expression of ORMDL3 by Genotype",
    x = "Genotype",
    y = "Expression Level"
  ) +
  theme_minimal()
```

```
Warning in geom_boxplot(aes(x = geno, y = exp, fill = geno, notch = TRUE)):
Ignoring unknown aesthetics: notch
```

